

5.Example txt

Name: P.Bindhu Madhavi

Roll Number: 2520030077

Step 1: Create database

The screenshot shows the MySQL Workbench interface. The left sidebar contains a 'MANAGEMENT' section with options like Server Status, Client Connections, Users and Privileges, Status and System, Data Export, and Data Import/Restore. Below this is an 'INSTANCE' section with Startup/Shutdown, Server Logs, and Options File. The 'PERFORMANCE' section includes Dashboard and Performance Report. The 'Administration' section is also visible. The main window displays a SQL script with three lines: 1. A comment line: `2520030077 - Bindhu Madhavi`; 2. `CREATE DATABASE IF NOT EXISTS joins_db;`; 3. `USE joins_db;`. The 'Output' tab at the bottom shows the execution results in a table format.

#	Time	Action	Message
1	10:34:16	CREATE DATABASE IF NOT EXISTS joins_db	1 row(s) affected, 1 warning(s): 1007 Can
2	10:34:16	USE joins_db	0 row(s) affected

Step 2: Create physician Table

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator: week2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Operations* My Example*

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Report

Administration

Information

No object selected

```
1 2520030077 - Bindhu Madhavi
2 CREATE DATABASE IF NOT EXISTS joins_db;
3 USE joins_db;
4 CREATE TABLE physician (
5     employeeid INT PRIMARY KEY,
6     name VARCHAR(100),
7     position VARCHAR(100),
8     ssn VARCHAR(20)
9 );
```

Output

Action Output

#	Time	Action	Message
1	10:34:16	CREATE DATABASE IF NOT EXISTS joins_db	1 row(s) affected, 1 warning(s): 1007 Ca
2	10:34:16	USE joins_db	0 row(s) affected
3	10:35:39	CREATE TABLE physician (employeeid INT PRIMARY K...	0 row(s) affected

Step 3: Insert Data into physician

MySQL Workbench

Bindhu Madhavi Local instanc...

File Edit View Query Database Server Tools Scripting Help

Navigator: week2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Operations* My Example* x

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdo
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Rep

Administration

Information

No object selected

```

4 CREATE TABLE physician (
5     employeeid INT PRIMARY KEY,
6     name VARCHAR(100),
7     position VARCHAR(100),
8     ssn VARCHAR(20)
9 );
10 INSERT INTO physician VALUES
11 (1,'Bindhu Madhavii','Staff Internist','111111111'),
12 (2,'Bindhu Madhavi','Attending Physician','222222222'),
13 (3,'Bindhu Madhavi Turk','Surgical Attending Physician','333333333'),
14 (4,'Percival Cox','Senior Attending Physician','444444444'),
15 (5,'Bob Kelso','Head Chief of Medicine','555555555'),
16 (6,'Todd Quinlan','Surgical Attending Physician','666666666'),
17 (7,'John Wen','Surgical Attending Physician','777777777'),
18 (8,'Keith Dudemeister','MD Resident','888888888'),
19 (9,'Molly Clock','Attending Psychiatrist','999999999');

```

Output

Action Output

#	Time	Action	Message
1	10:34:16	CREATE DATABASE IF NOT EXISTS joins_db	1 row(s) affected, 1 warning(s): 1007 C
2	10:34:16	USE joins_db	0 row(s) affected
3	10:35:39	CREATE TABLE physician (employeeid INT PRIMARY K...	0 row(s) affected
4	10:37:11	CREATE TABLE physician (employeeid INT PRIMARY K...	Error Code: 1050. Table 'physician' alre
5	10:38:06	INSERT INTO physician VALUES (1,'Bindhu Madhavii','Staff...	9 row(s) affected Records: 9 Duplicate

Step 4:Create the department Table

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator: week2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Operations* My Example* x SQLAdditio

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdo
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Rep

Administration Information

No object selected

```

9 );
10 INSERT INTO physician VALUES
11 (1,'Bindhu Madhavii','Staff Internist','111111111'),
12 (2,'Bindhu Madhavi','Attending Physician','222222222'),
13 (3,'Bindhu Madhavi Turk','Surgical Attending Physician','333333333'),
14 (4,'Percival Cox','Senior Attending Physician','444444444'),
15 (5,'Bob Kelso','Head Chief of Medicine','555555555'),
16 (6,'Todd Quinlan','Surgical Attending Physician','666666666'),
17 (7,'John Wen','Surgical Attending Physician','777777777'),
18 (8,'Keith Dudemeister','MD Resident','888888888'),
19 (9,'Molly Clock','Attending Psychiatrist','999999999');
20 CREATE TABLE department (
21     departmentid INT PRIMARY KEY,
22     name VARCHAR(100),
23     head INT
24 );

```

Output

Action Output

#	Time	Action	Message
1	10:34:16	CREATE DATABASE IF NOT EXISTS joins_db	1 row(s) affected, 1 warning(s): 1007 Can't creat
2	10:34:16	USE joins_db	0 row(s) affected
3	10:35:39	CREATE TABLE physician (employeeid INT PRIMARY KEY,...	0 row(s) affected
4	10:37:11	CREATE TABLE physician (employeeid INT PRIMARY KEY,...	Error Code: 1050. Table 'physician' already exist
5	10:38:06	INSERT INTO physician VALUES (1,'Bindhu Madhavii','Staff Int...	9 row(s) affected Records: 9 Duplicates: 0 War
6	10:41:08	CREATE TABLE department (departmentid INT PRIMARY K...	0 row(s) affected

Step 5: Insert Data into department

MySQL Workbench

Bindhu Madhavi Local instanc...

File Edit View Query Database Server Tools Scripting Help

Navigator: eek2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Operations* My Example* x SQLAdditions

MANAGEMENT

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INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Rep

Administration Information

No object selected

```

14 (4,'Percival Cox','Senior Attending Physician','444444444'),
15 (5,'Bob Kelso','Head Chief of Medicine','555555555'),
16 (6,'Todd Quinlan','Surgical Attending Physician','666666666'),
17 (7,'John Wen','Surgical Attending Physician','777777777'),
18 (8,'Keith Dudemeister','MD Resident','888888888'),
19 (9,'Molly Clock','Attending Psychiatrist','999999999');
20 CREATE TABLE department (
21     departmentid INT PRIMARY KEY,
22     name VARCHAR(100),
23     head INT
24 );
25 2520030077- Bindhu Madhavi
26 INSERT INTO department VALUES
27 (1,'General Medicine',4),
28 (2,'Surgery',7),
29 (3,'Psychiatry',9);

```

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Context Help Snippets

Output

Action Output

#	Time	Action	Message
1	10:34:16	CREATE DATABASE IF NOT EXISTS joins_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'jo
2	10:34:16	USE joins_db	0 row(s) affected
3	10:35:39	CREATE TABLE physician (employeeid INT PRIMARY KEY,...	0 row(s) affected
4	10:37:11	CREATE TABLE physician (employeeid INT PRIMARY KEY,...	Error Code: 1050. Table 'physician' already exists
5	10:38:06	INSERT INTO physician VALUES (1,'Bindhu Madhavi','Staff Int...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
6	10:41:08	CREATE TABLE department (departmentid INT PRIMARY K...	0 row(s) affected
7	10:42:11	INSERT INTO department VALUES (1,'General Medicine',4), (2,...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

Step 6: Inner Join

MySQL Workbench

Bindhu Madhavi Local instanc...

File Edit View Query Database Server Tools Scripting Help

Navigator: eek2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Operations* My Example* x SQLAdditions

MANAGEMENT

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INSTANCE

- Startup / Shutdo
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Rep

Administration

Information

No object selected

```

30 2520030077- Bindhu Madhavi
31 SELECT d.name AS Department,
32        p.name AS Head_Physician
33 FROM department d
34 JOIN physician p
35 ON d.head = p.employeeid;
36

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Department	Head_Physician
General Medicine	Percival Cox
Surgery	John Wen
Psychiatry	Molly Clock

Result Grid

Form Editor

Result 1 x Read Only Context Help

Output

Action Output

#	Time	Action	Message
1	10:34:16	CREATE DATABASE IF NOT EXISTS joins_db	1 row(s) affected, 1 warning(s): 1007 Can't create database
2	10:34:16	USE joins_db	0 row(s) affected
3	10:35:39	CREATE TABLE physician (employeeid INT PRIMARY ...	0 row(s) affected
4	10:37:11	CREATE TABLE physician (employeeid INT PRIMARY ...	Error Code: 1050. Table 'physician' already exists
5	10:38:06	INSERT INTO physician VALUES (1,'Bindhu Madhavi','Sta...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
6	10:41:08	CREATE TABLE department (departmentid INT PRIMA...	0 row(s) affected
7	10:42:11	INSERT INTO department VALUES (1,'General Medicine',4...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
8	10:43:34	SELECT d.name AS Department, p.name AS Head_Ph...	3 row(s) returned

Step 7: Where Clause

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator: week2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Operations* My Example* x SQLAd

MANAGEMENT

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INSTANCE

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- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Rep

Administration

Information

No object selected

```
32 p.name AS Head_Physician
33 FROM department d
34 JOIN physician p
35 ON d.head = p.employeeid;
36 SELECT *
37 FROM physician
38 WHERE position='Surgical Attending Physician';
```

Result Grid

	employeeid	name	position	ssn
▶	3	Bindhu Madhavi Turk	Surgical Attending Physician	333333333
	6	Todd Quinlan	Surgical Attending Physician	666666666
	7	John Wen	Surgical Attending Physician	777777777
*	NULL	NULL	NULL	NULL

physician 2 x Apply Context

Output

Action Output

#	Time	Action	Message
⚠	1 10:34:16	CREATE DATABASE IF NOT EXISTS joins_db	1 row(s) affected, 1 warning(s): 1007 Can't creat
✓	2 10:34:16	USE joins_db	0 row(s) affected
✓	3 10:35:39	CREATE TABLE physician (employeeid INT PRIMARY ...	0 row(s) affected
✗	4 10:37:11	CREATE TABLE physician (employeeid INT PRIMARY ...	Error Code: 1050. Table 'physician' already exist
✓	5 10:38:06	INSERT INTO physician VALUES (1,'Bindhu Madhavi','Sta...	9 row(s) affected Records: 9 Duplicates: 0 Wa
✓	6 10:41:08	CREATE TABLE department (departmentid INT PRIMA...	0 row(s) affected
✓	7 10:42:11	INSERT INTO department VALUES (1,'General Medicine',4...	3 row(s) affected Records: 3 Duplicates: 0 Wa
✓	8 10:43:34	SELECT d.name AS Department, p.name AS Head_Ph...	3 row(s) returned
✓	9 10:44:43	SELECT * FROM physician WHERE position='Surgical Atte...	3 row(s) returned

Step 8: Where + Like

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator: week2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Operations* My Example* x

MANAGEMENT

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- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Report

Administration

Information

No object selected

```
35 ON d.head = p.employeeid;
36 • SELECT *
37 FROM physician
38 WHERE position='Surgical Attending Physician';
39 • SELECT *
40 FROM physician
41 WHERE name LIKE 'Bindhu Madhavi%';
```

Limit to 1000 rows

Result Grid

	employeeid	name	position	ssn
▶	1	Bindhu Madhavi	Staff Internist	111111111
	2	Bindhu Madhavi	Attending Physician	222222222
	3	Bindhu Madhavi Turk	Surgical Attending Physician	333333333
*	NULL	NULL	NULL	NULL

physician 4 x

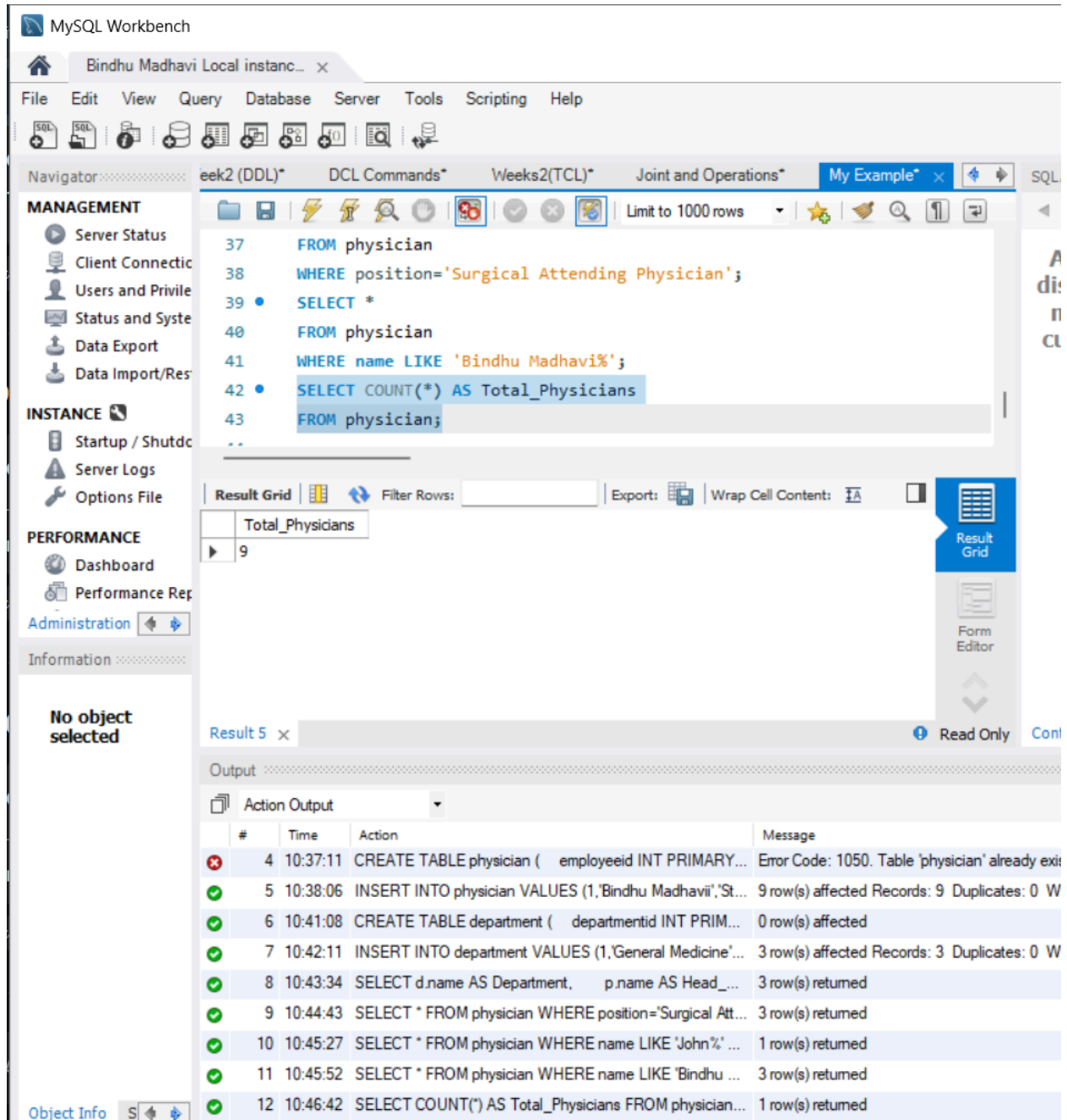
Output

Action Output

#	Time	Action	Message
✓	3 10:35:39	CREATE TABLE physician (employeeid INT PRIMARY...	0 row(s) affected
✗	4 10:37:11	CREATE TABLE physician (employeeid INT PRIMARY...	Error Code: 1050. Table 'physician' already exists
✓	5 10:38:06	INSERT INTO physician VALUES (1,'Bindhu Madhavi','St...	9 row(s) affected Records: 9 Duplicates: 0 V
✓	6 10:41:08	CREATE TABLE department (departmentid INT PRIM...	0 row(s) affected
✓	7 10:42:11	INSERT INTO department VALUES (1,'General Medicine'...	3 row(s) affected Records: 3 Duplicates: 0 V
✓	8 10:43:34	SELECT d.name AS Department, p.name AS Head_...	3 row(s) returned
✓	9 10:44:43	SELECT * FROM physician WHERE position='Surgical Att...	3 row(s) returned
✓	10 10:45:27	SELECT * FROM physician WHERE name LIKE 'John%' ...	1 row(s) returned
✓	11 10:45:52	SELECT * FROM physician WHERE name LIKE 'Bindhu ...	3 row(s) returned

Object Info

Step 9: Count()



The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
37 FROM physician
38 WHERE position='Surgical Attending Physician';
39 • SELECT *
40 FROM physician
41 WHERE name LIKE 'Bindhu Madhavi%';
42 • SELECT COUNT(*) AS Total_Physicians
43 FROM physician;
```

The query is executed, and the result is displayed in the 'Result Grid' tab. The result shows a single row with the value 9 for the column 'Total_Physicians'.

Total_Physicians
9

The 'Output' tab shows the 'Action Output' log, which includes the following entries:

#	Time	Action	Message
4	10:37:11	CREATE TABLE physician (employeeid INT PRIMARY...	Error Code: 1050. Table 'physician' already exists
5	10:38:06	INSERT INTO physician VALUES (1,'Bindhu Madhavi','St...	9 row(s) affected Records: 9 Duplicates: 0 W
6	10:41:08	CREATE TABLE department (departmentid INT PRIM...	0 row(s) affected
7	10:42:11	INSERT INTO department VALUES (1,'General Medicine'...	3 row(s) affected Records: 3 Duplicates: 0 W
8	10:43:34	SELECT d.name AS Department, p.name AS Head_...	3 row(s) returned
9	10:44:43	SELECT * FROM physician WHERE position='Surgical Att...	3 row(s) returned
10	10:45:27	SELECT * FROM physician WHERE name LIKE 'John%' ...	1 row(s) returned
11	10:45:52	SELECT * FROM physician WHERE name LIKE 'Bindhu ...	3 row(s) returned
12	10:46:42	SELECT COUNT(*) AS Total_Physicians FROM physician...	1 row(s) returned

Step 10: Count Distinct

MySQL Workbench

Bindhu Madhavi Local instanc... x

File Edit View Query Database Server Tools Scripting Help

Navigator: week2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Operations* My Example* x

MANAGEMENT

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- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdo
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Rep

Administration

Information

No object selected

```
39 • SELECT *
40 FROM physician
41 WHERE name LIKE 'Bindhu Madhavi%';
42 • SELECT COUNT(*) AS Total_Physicians
43 FROM physician;
44 • SELECT COUNT(DISTINCT position)
45 FROM physician;
```

Result Grid

	COUNT(DISTINCT position)
▶	7

Result 6 x

Output

Action Output

#	Time	Action	Message
✓ 5	10:38:06	INSERT INTO physician VALUES (1,'Bindhu Madhavi','St...	9 row(s) affected Records: 9 Duplica
✓ 6	10:41:08	CREATE TABLE department (departmentid INT PRIM...	0 row(s) affected
✓ 7	10:42:11	INSERT INTO department VALUES (1,'General Medicine'...	3 row(s) affected Records: 3 Duplica
✓ 8	10:43:34	SELECT d.name AS Department, p.name AS Head_...	3 row(s) returned
✓ 9	10:44:43	SELECT * FROM physician WHERE position='Surgical Att...	3 row(s) returned
✓ 10	10:45:27	SELECT * FROM physician WHERE name LIKE 'John%' ...	1 row(s) returned
✓ 11	10:45:52	SELECT * FROM physician WHERE name LIKE 'Bindhu ...	3 row(s) returned
✓ 12	10:46:42	SELECT COUNT(*) AS Total_Physicians FROM physician...	1 row(s) returned
✓ 13	10:47:25	SELECT COUNT(DISTINCT position) FROM physician LI...	1 row(s) returned

Step 11: Group By

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator: eek2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Operations* My Example* x

MANAGEMENT

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- Client Connections
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- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Report

Administration Information

No object selected

```
44 SELECT COUNT(DISTINCT position)
45 FROM physician;
46 2520030077- Bindhu Madhavi
47 SELECT position,
48 COUNT(*) AS Total_Employees
49 FROM physician
50 GROUP BY position;
```

Result Grid

position	Total_Employees
Staff Internist	1
Attending Physician	1
Surgical Attending Physician	3
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

Result 7 x Read Only

Output

Action Output

#	Time	Action	Message
6	10:41:08	CREATE TABLE department (departmentid INT PRIM...	0 row(s) affected
7	10:42:11	INSERT INTO department VALUES (1,'General Medicine'...	3 row(s) affected Records: 3 Duplicates
8	10:43:34	SELECT d.name AS Department, p.name AS Head_...	3 row(s) returned
9	10:44:43	SELECT * FROM physician WHERE position='Surgical Att...	3 row(s) returned
10	10:45:27	SELECT * FROM physician WHERE name LIKE 'John%' ...	1 row(s) returned
11	10:45:52	SELECT * FROM physician WHERE name LIKE 'Bindhu ...	3 row(s) returned
12	10:46:42	SELECT COUNT(*) AS Total_Physicians FROM physician...	1 row(s) returned
13	10:47:25	SELECT COUNT(DISTINCT position) FROM physician LI...	1 row(s) returned
14	10:48:12	SELECT position, COUNT(*) AS Total_Employees FR...	7 row(s) returned

Step 12: Group By + Having

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
2520030077- Bindhu Madhavi
SELECT position,
       COUNT(*) AS Total_Employees
FROM physician
GROUP BY position
HAVING COUNT(*) > 1;
```

The query is highlighted in blue. The result grid below the query shows one row:

position	Total_Employees
Surgical Attending Physician	3

The output pane at the bottom shows the execution log. The last two entries (15 and 16) show an error:

#	Time	Action	Message
15	10:49:32	FROM physician GROUP BY position HAVING COUNT(*) ...	Error Code: 1064. You have an error in your SQ
16	10:49:47	FROM physician GROUP BY position HAVING COUNT(*) ...	Error Code: 1064. You have an error in your SQ

Step 13: Order By

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator: week2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Operations* My Example* x

MANAGEMENT

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INSTANCE

- Startup / Shutdown
- Server Logs
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PERFORMANCE

- Dashboard
- Performance Report

Administration

Information

No object selected

```

54
55 FROM physician
56 GROUP BY position
57 HAVING COUNT(*) > 1;
58 SELECT *
59 FROM physician
60 ORDER BY name ASC;

```

Result Grid

employeeid	name	position	ssn
2	Bindhu Madhavi	Attending Physician	222222222
3	Bindhu Madhavi Turk	Surgical Attending Physician	333333333
1	Bindhu Madhavi	Staff Internist	111111111
5	Bob Kelso	Head Chief of Medicine	555555555
7	John Wen	Surgical Attending Physician	777777777
8	Keith Dudemeister	MD Resident	888888888
9	Molly Clock	Attending Psychiatrist	999999999
4	Percival Cox	Senior Attending Physician	444444444

physician 9 x

Output

Action Output

#	Time	Action	Message
✓ 10	10:45:27	SELECT * FROM physician WHERE name LIKE 'John%' ...	1 row(s) returned
✓ 11	10:45:52	SELECT * FROM physician WHERE name LIKE 'Bindhu ...	3 row(s) returned
✓ 12	10:46:42	SELECT COUNT(*) AS Total_Physicians FROM physician...	1 row(s) returned
✓ 13	10:47:25	SELECT COUNT(DISTINCT position) FROM physician LI...	1 row(s) returned
✓ 14	10:48:12	SELECT position, COUNT(*) AS Total_Employees FR...	7 row(s) returned
✗ 15	10:49:32	FROM physician GROUP BY position HAVING COUNT(*) ...	Error Code: 1064. You have an error in your S
✗ 16	10:49:47	FROM physician GROUP BY position HAVING COUNT(*) ...	Error Code: 1064. You have an error in your S
✓ 17	10:49:58	SELECT position, COUNT(*) AS Total_Employees ...	1 row(s) returned
✓ 18	10:50:49	SELECT * FROM physician ORDER BY name ASC LIMIT...	9 row(s) returned

Step 14: Order By Desc

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator: week2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Operations* My Example* x SQLA

MANAGEMENT

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INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Report

Administration Administration

Information: No object selected

```
57 HAVING COUNT(*) > 1;
58 SELECT *
59 FROM physician
60 ORDER BY name ASC;
61 SELECT *
62 FROM physician
63 ORDER BY employeeid DESC;
```

Result Grid

	employeeid	name	position	ssn
6		Todd Quinlan	Surgical Attending Physician	666666666
5		Bob Kelso	Head Chief of Medicine	555555555
4		Percival Cox	Senior Attending Physician	444444444
3		Bindhu Madhavi Turk	Surgical Attending Physician	333333333
2		Bindhu Madhavi	Attending Physician	222222222
1		Bindhu Madhavi	Staff Internist	111111111
*	NULL	NULL	NULL	NULL

physician 10 x Apply Conte

Output

Action Output

#	Time	Action	Message
✓ 11	10:45:52	SELECT * FROM physician WHERE name LIKE 'Bindhu ...	3 row(s) returned
✓ 12	10:46:42	SELECT COUNT(*) AS Total_Physicians FROM physician...	1 row(s) returned
✓ 13	10:47:25	SELECT COUNT(DISTINCT position) FROM physician LI...	1 row(s) returned
✓ 14	10:48:12	SELECT position, COUNT(*) AS Total_Employees FR...	7 row(s) returned
✗ 15	10:49:32	FROM physician GROUP BY position HAVING COUNT(*) ...	Error Code: 1064. You have an error in your SQL
✗ 16	10:49:47	FROM physician GROUP BY position HAVING COUNT(*) ...	Error Code: 1064. You have an error in your SQL
✓ 17	10:49:58	SELECT position, COUNT(*) AS Total_Employees ...	1 row(s) returned
✓ 18	10:50:49	SELECT * FROM physician ORDER BY name ASC LIMIT...	9 row(s) returned
✓ 19	10:51:36	SELECT * FROM physician ORDER BY employeeid DES...	9 row(s) returned

Object Info

Step 15: Where + In

MySQL Workbench

The screenshot shows the MySQL Workbench interface for a local instance named 'Bindhu Madhavi'. The left sidebar contains navigation panels for 'MANAGEMENT', 'INSTANCE', and 'PERFORMANCE'. The main editor window displays a SQL query in the 'My Example' tab:

```
64 2520030077- Bindhu Madhavi
65 SELECT *
66 FROM physician
67 WHERE employeeid IN
68 (
69     SELECT head
70     FROM department
```

Below the query editor, the 'Result Grid' is visible, showing a table with 4 columns: employeeid, name, position, and ssn. The data includes three rows of physician information and a row of NULL values.

employeeid	name	position	ssn
4	Percival Cox	Senior Attending Physician	44444444
7	John Wen	Surgical Attending Physician	77777777
9	Molly Clock	Attending Psychiatrist	99999999
NULL	NULL	NULL	NULL

The bottom panel shows the 'Output' tab with 'Action Output' selected, displaying a log of SQL actions and their results. The log includes successful queries and two error messages (Error Code: 1064).

#	Time	Action	Message
12	10:46:42	SELECT COUNT(*) AS Total_Physicians FROM physician...	1 row(s) returned
13	10:47:25	SELECT COUNT(DISTINCT position) FROM physician LI...	1 row(s) returned
14	10:48:12	SELECT position, COUNT(*) AS Total_Employees FR...	7 row(s) returned
15	10:49:32	FROM physician GROUP BY position HAVING COUNT(*) ...	Error Code: 1064. You have an error i
16	10:49:47	FROM physician GROUP BY position HAVING COUNT(*) ...	Error Code: 1064. You have an error i
17	10:49:58	SELECT position, COUNT(*) AS Total_Employees ...	1 row(s) returned
18	10:50:49	SELECT * FROM physician ORDER BY name ASC LIMIT...	9 row(s) returned
19	10:51:36	SELECT * FROM physician ORDER BY employeeid DES...	9 row(s) returned
20	10:52:31	SELECT * FROM physician WHERE employeeid IN (S...	3 row(s) returned

Step 16: Subquery (Not In)

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
72 SELECT *
73 FROM physician
74 WHERE employeeid NOT IN
75 (
76     SELECT head
77     FROM department
78 );
```

The 'Result Grid' tab is active, displaying the following data:

employeeid	name	position	ssn
1	Bindhu Madhavii	Staff Internist	111111111
2	Bindhu Madhavi	Attending Physician	222222222
3	Bindhu Madhavi Turk	Surgical Attending Physician	333333333
5	Bob Kelso	Head Chief of Medicine	555555555
6	Todd Quinlan	Surgical Attending Physician	666666666
8	Keith Dudemeister	MD Resident	888888888
*	NULL	NULL	NULL

The 'Output' tab shows a list of recent SQL actions and their results:

#	Time	Action	Message
✓ 13	10:47:25	SELECT COUNT(DISTINCT position) FROM physician LI...	1 row(s) returned
✓ 14	10:48:12	SELECT position, COUNT(*) AS Total_Employees FR...	7 row(s) returned
✗ 15	10:49:32	FROM physician GROUP BY position HAVING COUNT(*) ...	Error Code: 1064. You have an error in you
✗ 16	10:49:47	FROM physician GROUP BY position HAVING COUNT(*) ...	Error Code: 1064. You have an error in you
✓ 17	10:49:58	SELECT position, COUNT(*) AS Total_Employees ...	1 row(s) returned
✓ 18	10:50:49	SELECT * FROM physician ORDER BY name ASC LIMIT...	9 row(s) returned
✓ 19	10:51:36	SELECT * FROM physician ORDER BY employeeid DES...	9 row(s) returned
✓ 20	10:52:31	SELECT * FROM physician WHERE employeeid IN (S...	3 row(s) returned
✓ 21	10:53:47	SELECT * FROM physician WHERE employeeid NOT IN ...	6 row(s) returned

Step 17: Group By + Order By

MySQL Workbench interface showing a query execution session.

Navigator: Bindhu Madhavi Local instanc... x

File Edit View Query Database Server Tools Scripting Help

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdo
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Rep

Administration

Information

No object selected

Query Editor:

```
78 );
79 2520030077- Bindhu Madhavi
80 SELECT position,
81     COUNT(*) AS Total_Employees
82 FROM physician
83 GROUP BY position
84 ORDER BY Total_Employees DESC;
```

Result Grid:

position	Total_Employees
Surgical Attending Physician	3
Staff Internist	1
Attending Physician	1
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

Output:

Action Output

#	Time	Action	Message
✓ 14	10:48:12	SELECT position, COUNT(*) AS Total_Employees FR...	7 row(s) returned
✗ 15	10:49:32	FROM physician GROUP BY position HAVING COUNT(*) ...	Error Code: 1064. You have an error in your SQL
✗ 16	10:49:47	FROM physician GROUP BY position HAVING COUNT(*) ...	Error Code: 1064. You have an error in your SQL
✓ 17	10:49:58	SELECT position, COUNT(*) AS Total_Employees ...	1 row(s) returned
✓ 18	10:50:49	SELECT * FROM physician ORDER BY name ASC LIMIT...	9 row(s) returned
✓ 19	10:51:36	SELECT * FROM physician ORDER BY employeeid DES...	9 row(s) returned
✓ 20	10:52:31	SELECT * FROM physician WHERE employeeid IN (S...	3 row(s) returned
✓ 21	10:53:47	SELECT * FROM physician WHERE employeeid NOT IN ...	6 row(s) returned
✓ 22	10:55:14	SELECT position, COUNT(*) AS Total_Employees FR...	7 row(s) returned

Step 18: Where + Like

MySQL Workbench

Bindhu Madhavi Local instanc...

File Edit View Query Database Server Tools Scripting Help

Navigator: week2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Operations* My Example* x SQLAdd

MANAGEMENT

- Server Status
- Client Connecti...
- Users and Privile...
- Status and Syste...
- Data Export
- Data Import/Res...

INSTANCE

- Startup / Shutdc...
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Rep...

Administration

Information: No object selected

```
81      COUNT(*) AS Total_Employees
82  FROM physician
83  GROUP BY position
84  ORDER BY Total_Employees DESC;
85  SELECT *
86  FROM physician
87  WHERE position LIKE '%Attending%';
```

Result Grid

	employeeid	name	position	ssn
▶	2	Bindhu Madhavi	Attending Physician	222222222
	3	Bindhu Madhavi Turk	Surgical Attending Physician	333333333
	4	Percival Cox	Senior Attending Physician	444444444
	6	Todd Quinlan	Surgical Attending Physician	666666666
	7	John Wen	Surgical Attending Physician	777777777
	9	Molly Clock	Attending Psychiatrist	999999999
*	NULL	NULL	NULL	NULL

physician 14 x Apply Context

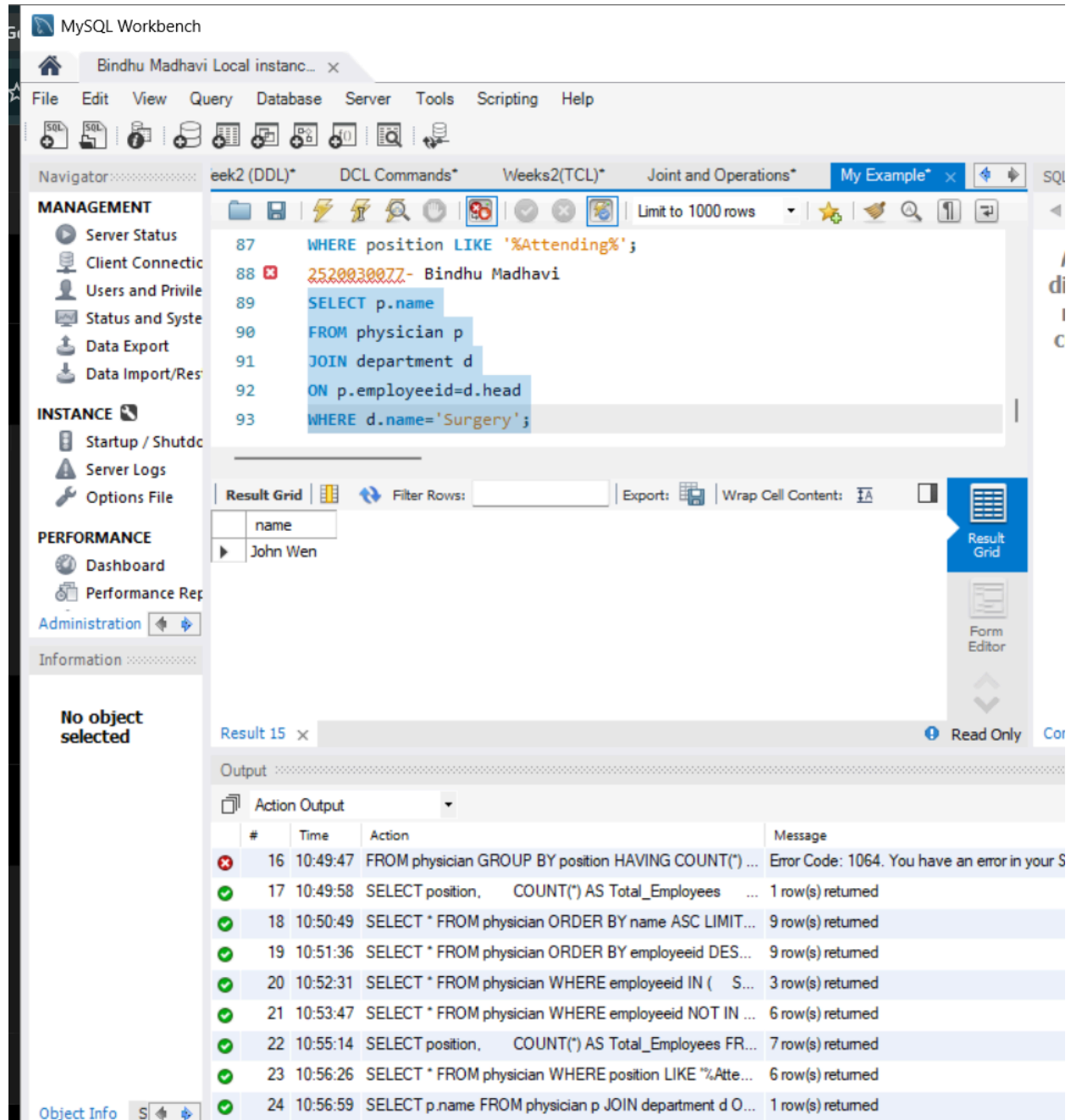
Output

Action Output

#	Time	Action	Message
✗ 15	10:49:32	FROM physician GROUP BY position HAVING COUNT(*) ...	Error Code: 1064. You have an error in your SQL s...
✗ 16	10:49:47	FROM physician GROUP BY position HAVING COUNT(*) ...	Error Code: 1064. You have an error in your SQL s...
✓ 17	10:49:58	SELECT position, COUNT(*) AS Total_Employees ...	1 row(s) returned
✓ 18	10:50:49	SELECT * FROM physician ORDER BY name ASC LIMIT ...	9 row(s) returned
✓ 19	10:51:36	SELECT * FROM physician ORDER BY employeeid DES...	9 row(s) returned
✓ 20	10:52:31	SELECT * FROM physician WHERE employeeid IN (S...	3 row(s) returned
✓ 21	10:53:47	SELECT * FROM physician WHERE employeeid NOT IN ...	6 row(s) returned
✓ 22	10:55:14	SELECT position, COUNT(*) AS Total_Employees FR...	7 row(s) returned
✓ 23	10:56:26	SELECT * FROM physician WHERE position LIKE "%Atte...	6 row(s) returned

Object Info

Step 19: Join + Where



The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' section with options like Server Status, Client Connections, Users and Privileges, Status and System, Data Export, and Data Import/Restore. The 'INSTANCE' section includes Startup/Shutdown, Server Logs, and Options File. The 'PERFORMANCE' section includes Dashboard and Performance Report. The 'Administration' and 'Information' sections are also visible.

The main window displays a SQL query in the 'My Example' tab:

```
87 WHERE position LIKE '%Attending%';
88 2520030077- Bindhu Madhavi
89 SELECT p.name
90 FROM physician p
91 JOIN department d
92 ON p.employeeid=d.head
93 WHERE d.name='Surgery';
```

The 'Result Grid' shows a single row with the name 'John Wen'.

The 'Action Output' window shows the results of the query, including the error message 'Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '' at line 1'.

#	Time	Action	Message
16	10:49:47	FROM physician GROUP BY position HAVING COUNT(*) ...	Error Code: 1064. You have an error in your S
17	10:49:58	SELECT position, COUNT(*) AS Total_Employees ...	1 row(s) returned
18	10:50:49	SELECT * FROM physician ORDER BY name ASC LIMIT ...	9 row(s) returned
19	10:51:36	SELECT * FROM physician ORDER BY employeeid DES...	9 row(s) returned
20	10:52:31	SELECT * FROM physician WHERE employeeid IN (S...	3 row(s) returned
21	10:53:47	SELECT * FROM physician WHERE employeeid NOT IN ...	6 row(s) returned
22	10:55:14	SELECT position, COUNT(*) AS Total_Employees FR...	7 row(s) returned
23	10:56:26	SELECT * FROM physician WHERE position LIKE "%Atte...	6 row(s) returned
24	10:56:59	SELECT p.name FROM physician p JOIN department d O...	1 row(s) returned

Step 20: Join + Order By

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator: week2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Operations* My Example* x

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Report

Administration Information

No object selected

```
94 2520030077- Bindhu Madhavi
95 SELECT d.name,
96        p.name
97 FROM department d
98 JOIN physician p
99 ON d.head=p.employeeid
100 ORDER BY d.name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	name	name
▶	General Medicine	Percival Cox
	Psychiatry	Molly Clock
	Surgery	John Wen

Result 17 x Read Only

Output

Action Output

#	Time	Action	Message
✓ 18	10:50:49	SELECT * FROM physician ORDER BY name ASC LIMIT...	9 row(s) returned
✓ 19	10:51:36	SELECT * FROM physician ORDER BY employeeid DES...	9 row(s) returned
✓ 20	10:52:31	SELECT * FROM physician WHERE employeeid IN (S...	3 row(s) returned
✓ 21	10:53:47	SELECT * FROM physician WHERE employeeid NOT IN ...	6 row(s) returned
✓ 22	10:55:14	SELECT position, COUNT(*) AS Total_Employees FR...	7 row(s) returned
✓ 23	10:56:26	SELECT * FROM physician WHERE position LIKE "%Atte...	6 row(s) returned
✓ 24	10:56:59	SELECT p.name FROM physician p JOIN department d O...	1 row(s) returned
✓ 25	10:58:48	SELECT d.name, p.name FROM department d JOIN ...	3 row(s) returned
✓ 26	10:59:04	SELECT d.name, p.name FROM department d JOIN ...	3 row(s) returned

Object Info

Step 21: Where + Not Like

MySQL Workbench

Bindhu Madhavi Local instanc... x

File Edit View Query Database Server Tools Scripting Help

Navigator: week2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Operations* My Example* x SQLA

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Report

Administration Administration

Information No object selected

```
97 FROM department d
98 JOIN physician p
99 ON d.head=p.employeeid
100 ORDER BY d.name;
101 SELECT *
102 FROM physician
103 WHERE position NOT LIKE '%Surgical%';
```

Result Grid

	employeeid	name	position	ssn
1	1	Bindhu Madhavi	Staff Internist	111111111
2	2	Bindhu Madhavi	Attending Physician	222222222
4	4	Percival Cox	Senior Attending Physician	444444444
5	5	Bob Kelso	Head Chief of Medicine	555555555
8	8	Keith Dudemeister	MD Resident	888888888
9	9	Molly Clock	Attending Psychiatrist	999999999
*	NULL	NULL	NULL	NULL

physician 18 x Apply Conte

Output

Action Output

#	Time	Action	Message
✓ 19	10:51:36	SELECT * FROM physician ORDER BY employeeid DES...	9 row(s) returned
✓ 20	10:52:31	SELECT * FROM physician WHERE employeeid IN (S...	3 row(s) returned
✓ 21	10:53:47	SELECT * FROM physician WHERE employeeid NOT IN ...	6 row(s) returned
✓ 22	10:55:14	SELECT position, COUNT(*) AS Total_Employees FR...	7 row(s) returned
✓ 23	10:56:26	SELECT * FROM physician WHERE position LIKE "%Atte...	6 row(s) returned
✓ 24	10:56:59	SELECT p.name FROM physician p JOIN department d O...	1 row(s) returned
✓ 25	10:58:48	SELECT d.name, p.name FROM department d JOIN ...	3 row(s) returned
✓ 26	10:59:04	SELECT d.name, p.name FROM department d JOIN ...	3 row(s) returned
✓ 27	10:59:44	SELECT * FROM physician WHERE position NOT LIKE '...	6 row(s) returned

Step 22: Count + Having

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator: week2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Operations* My Example* x SQLAdd

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Report

Administration Information

No object selected

```
103 WHERE position NOT LIKE '%Surgical%';
104 2520030077- Bindhu Madhavi
105 SELECT position,
106 COUNT(*) AS Total
107 FROM physician
108 GROUP BY position
109 HAVING COUNT(*) >= 2;
```

Result Grid

position	Total
Surgical Attending Physician	3

Result 19 x Read Only Context

Output

Action Output

#	Time	Action	Message
20	10:52:31	SELECT * FROM physician WHERE employeeid IN (S...	3 row(s) returned
21	10:53:47	SELECT * FROM physician WHERE employeeid NOT IN ...	6 row(s) returned
22	10:55:14	SELECT position, COUNT(*) AS Total_Employees FR...	7 row(s) returned
23	10:56:26	SELECT * FROM physician WHERE position LIKE "%Atte...	6 row(s) returned
24	10:56:59	SELECT p.name FROM physician p JOIN department d O...	1 row(s) returned
25	10:58:48	SELECT d.name, p.name FROM department d JOIN ...	3 row(s) returned
26	10:59:04	SELECT d.name, p.name FROM department d JOIN ...	3 row(s) returned
27	10:59:44	SELECT * FROM physician WHERE position NOT LIKE '...	6 row(s) returned
28	11:00:27	SELECT position, COUNT(*) AS Total FROM physici...	1 row(s) returned

Step 23: Exists

MySQL Workbench

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
109 HAVING COUNT(*) >= 2;  
110 2520030077- Bindhu Madhavi  
111 SELECT *  
112 FROM department d  
113 WHERE EXISTS  
114 (  
115 SELECT 1
```

The Result Grid shows the following data:

departmentid	name	head
1	General Medicine	4
2	Surgery	7
3	Psychiatry	9
NULL	NULL	NULL

The Action Output pane shows the following results:

#	Time	Action	Message
21	10:53:47	SELECT * FROM physician WHERE employeeid NOT IN ...	6 row(s) returned
22	10:55:14	SELECT position, COUNT(*) AS Total_Employees FR...	7 row(s) returned
23	10:56:26	SELECT * FROM physician WHERE position LIKE "%Atte...	6 row(s) returned
24	10:56:59	SELECT p.name FROM physician p JOIN department d O...	1 row(s) returned
25	10:58:48	SELECT d.name, p.name FROM department d JOIN ...	3 row(s) returned
26	10:59:04	SELECT d.name, p.name FROM department d JOIN ...	3 row(s) returned
27	10:59:44	SELECT * FROM physician WHERE position NOT LIKE '...	6 row(s) returned
28	11:00:37	SELECT position, COUNT(*) AS Total FROM physicia...	1 row(s) returned
29	11:01:44	SELECT * FROM department d WHERE EXISTS (SEL...	3 row(s) returned

Step 24: Max()

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator: eek2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Operations* My Example* x SQLAdditions

MANAGEMENT

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- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Rep

Administration

Information

No object selected

```

115 SELECT 1
116 FROM physician p
117 WHERE p.employeeid=d.head
118 );
119 2520030077- Bindhu Madhavi
120 SELECT MAX(employeeid)
121 FROM physician;

```

Result Grid

MAX(employeeid)
9

Result 21 x Read Only Context Help

Output

Action Output

#	Time	Action	Message
✓ 22	10:55:14	SELECT position, COUNT(*) AS Total_Employees FR...	7 row(s) returned
✓ 23	10:56:26	SELECT * FROM physician WHERE position LIKE "%Atte...	6 row(s) returned
✓ 24	10:56:59	SELECT p.name FROM physician p JOIN department d O...	1 row(s) returned
✓ 25	10:58:48	SELECT d.name, p.name FROM department d JOIN ...	3 row(s) returned
✓ 26	10:59:04	SELECT d.name, p.name FROM department d JOIN ...	3 row(s) returned
✓ 27	10:59:44	SELECT * FROM physician WHERE position NOT LIKE '...	6 row(s) returned
✓ 28	11:00:37	SELECT position, COUNT(*) AS Total FROM physicia...	1 row(s) returned
✓ 29	11:01:44	SELECT * FROM department d WHERE EXISTS (SEL...	3 row(s) returned
✓ 30	11:03:00	SELECT MAX(employeeid) FROM physician LIMIT 0, 1000	1 row(s) returned

Object Info

Step 25: Min()

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator: week2 (DDL)* DCL Commands* Weeks2(TCL)* Joint and Operations* My Example* x

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Rep

Administration

Information

No object selected

```
117 WHERE p.employeeid=d.head
118 );
119 2520030077- Bindhu Madhavi
120 SELECT MAX(employeeid)
121 FROM physician;
122 SELECT MIN(employeeid)
123 FROM physician;
```

Result Grid

MIN(employeeid)
1

Result 22 x Read Only

Output

Action Output

#	Time	Action	Message
✓ 23	10:56:26	SELECT * FROM physician WHERE position LIKE "%Atte...	6 row(s) returned
✓ 24	10:56:59	SELECT p.name FROM physician p JOIN department d O...	1 row(s) returned
✓ 25	10:58:48	SELECT d.name, p.name FROM department d JOIN ...	3 row(s) returned
✓ 26	10:59:04	SELECT d.name, p.name FROM department d JOIN ...	3 row(s) returned
✓ 27	10:59:44	SELECT * FROM physician WHERE position NOT LIKE '...	6 row(s) returned
✓ 28	11:00:37	SELECT position, COUNT(*) AS Total FROM physicia...	1 row(s) returned
✓ 29	11:01:44	SELECT * FROM department d WHERE EXISTS (SEL...	3 row(s) returned
✓ 30	11:03:00	SELECT MAX(employeeid) FROM physician LIMIT 0, 1000	1 row(s) returned
✓ 31	11:03:44	SELECT MIN(employeeid) FROM physician LIMIT 0, 1000	1 row(s) returned

Object Info